

**Dataset Explanations for Assignment**  
Course Title: Artificial Intelligence and Machine Learning Lab  
Course Code: STA 0542 5184  
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### 01. Credit Data

The Credit dataset is a well-known benchmark dataset designed to study consumer credit behaviour, especially how demographic and financial characteristics relate to credit card usage and debt. It originally comes from real-world consumer credit data but has been cleaned, anonymized, and standardized for analysis purposes.

| Variables Name | Variables Description                                                                                                                                      | Variables Type |
|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| Income         | Annual income of the individual (in thousands of dollars)                                                                                                  | Numeric        |
| Limit          | Credit limit assigned to the individual on their credit card(s)                                                                                            | Numeric        |
| Ratings        | A credit score or internal rating assigned to the individual by the bank or financial institution. Higher values usually indicate better credit worthiness | Numeric        |
| Cards          | Number of credit cards the individual owns                                                                                                                 | Numeric        |
| Age            | Age of the individual (in years)                                                                                                                           | Numeric        |
| Education      | Years of formal education completed                                                                                                                        | Numeric        |
| Own            | Whether the individual owns a house/property                                                                                                               | Categorical    |
| Student        | Whether the individual is currently a student                                                                                                              | Categorical    |
| Married        | Marital status                                                                                                                                             | Categorical    |
| Region         | Geographical region of residence (e.g., South, West)                                                                                                       | Categorical    |
| Balance        | Current credit card balance                                                                                                                                | Numeric        |

### 02. Iris Data

The Iris dataset consists of 150 samples of iris flowers from three different species: *Setosa*, *Versicolor*, and *Virginica*. Each sample includes four features: *sepal length*, *sepal width*, *petal length*, and *petal width*. It was introduced by the British biologist and statistician Ronald Fisher in 1936 as an example of discriminant analysis. The dataset have the following format:

| Name            | Description                                                                                                        |
|-----------------|--------------------------------------------------------------------------------------------------------------------|
| Iris-setosa     | Characterized by its relatively small size, with distinctive characteristics in sepal and petal dimensions         |
| Iris-versicolor | Moderate in size, with features falling between those of setosa and virginica                                      |
| Iris-virginica  | Generally larger in size, with notable differences in sepal and petal dimensions compared to the other two species |
| Sepal Length    | The length of the iris flower's sepals (the green leaf-like structures that encase the flower bud)                 |
| Sepal Width     | The width of the iris flower's sepals                                                                              |
| Petal Length    | The length of the iris flower's petals (the colored structures of the flower)                                      |
| Petal Width     | The width of the iris flower's petals                                                                              |